Spreadsheet Basics and Presentation Design

Experiments 5 and 6

Discrete Mathematical Structures / Computer Applications

Artificial Intelligence and Machine Learning – Semester 1

Spreadsheet Basics

What is a Spreadsheet?

A **spreadsheet** is a computer application used to:

- Enter data
- Organize data
- Perform calculations
- Work with data in a table-like format

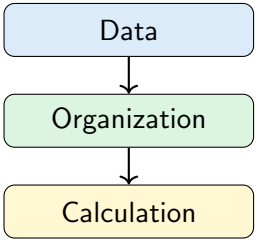
Examples of spreadsheet software:

- Microsoft Excel
- LibreOffice Calc
- Google Sheets

Where Are Spreadsheets Used?

Common uses:

- Student marks
- Attendance
- Monthly expenses
- Employee salaries
- Product lists
- Examination results



Basic Parts of a Spreadsheet

A spreadsheet is organized as a grid of:

- **Rows**
- **Columns**
- **Cells**

	A	B	C	D
1				
2				
3				
4				

- Rows are identified by numbers.
- Columns are identified by letters.
- The intersection of a row and column is a cell.

What is a Cell?

Definition

A **cell** is the intersection of a row and a column.

	A	B	C
1			
2	25	40	35
3			

Examples:

- 25 is stored in cell **A2**.
- 40 is stored in cell **B2**.
- 35 is stored in cell **C2**.

Cell Address

Definition

A **cell address** identifies the location of a cell.

$$\boxed{\text{Cell Address} = \text{Column Letter} + \text{Row Number}}$$

Examples:

A1 B2 C5 D10 E7

Remember

The column comes first and the row number comes second.

Classroom Activity – Identify the Cell

Consider the following spreadsheet:

	A	B	C
1	Name	Mark	Result
2	Arun	75	Pass
3	Anu	82	Pass
4	Rahul	68	Pass

Answer these questions:

1. What is the address of Arun?
2. What is the address of 82?
3. What is the address of “Pass” in Rahul’s row?

What is a Formula?

Definition

A **formula** is an expression used to perform a calculation in a spreadsheet.

Important: A formula normally begins with the symbol

$$\boxed{=}$$

Example 1:

$$= 10 + 20$$

$$\boxed{30}$$

Example 2 – Using Cell References
Suppose:

$$A1 = 10, \qquad B1 = 20$$

Cell Referencing

Definition

Cell referencing means using the address of a cell inside a formula.

Suppose the spreadsheet contains:

Cell	Value
A1	10
B1	20

The formula

$$= A1 + B1$$

performs:

$$10 + 20 = 30$$

Therefore:

Why Use Cell References?

Suppose:

Cell	Value
A1	10
B1	20

Formula:

$$= A1 + B1$$

Initially:

$$10 + 20 = 30$$

If A1 is changed to 50:

$$50 + 20 = 70$$

Understanding a Cell Range

What is a Range?

A **range** is a group of cells selected together.

For example:

$$\boxed{B2 : B4}$$

means:

$$B2, \quad B3, \quad B4$$

The colon symbol

$$:$$

means **“through”** or **“up to”**.

Therefore:

AVERAGE Function

Purpose

The **AVERAGE** function calculates the arithmetic mean of a group of numbers.

Example:
Marks:

$$50, \quad 60, \quad 70$$

Step 1 – Find the total

$$50 + 60 + 70 = 180$$

Step 2 – Count the values

$$3 \text{ values}$$

Step 3 – Calculate the average

$$\frac{180}{60}$$

COUNT Function

Purpose

The **COUNT** function counts the number of cells containing numeric values.

Consider:

Cell	Value
A1	25
A2	40
A3	60
A4	Hello
A5	75

Formula:

$$= COUNT(A1 : A5)$$

Numeric values are:

COUNTIF Function

Purpose

The **COUNTIF** function counts cells that satisfy a specified condition.

Consider:

Student	Result
Arun	Pass
Anu	Fail
Rahul	Pass
Meera	Pass
Amal	Fail

We want to count the number of students who obtained **Pass**.

Formula:

$$= COUNTIF(B2 : B6, \text{"Pass"})$$

The word "Pass" occurs:

COUNTIF Function

Purpose

The **COUNTIF** function counts cells that satisfy a specified condition.

Think of it as:

$$\boxed{\text{COUNT} + \text{IF condition}}$$

Example:

Student	Result
Arun	Pass
Anu	Fail
Rahul	Pass
Meera	Pass
Amal	Fail

Formula:

IF Function

Purpose

The **IF** function makes a decision based on a condition.

Basic syntax:

$$\boxed{= IF(condition, value_if_true, value_if_false)}$$

Example:

- Marks $\geq 40 \rightarrow$ Pass
- Marks $< 40 \rightarrow$ Fail

If the mark is in B2:

$$= IF(B2 >= 40, "Pass", "Fail")$$

Understanding IF Step-by-Step

Suppose:

$$B2 = 50$$

Formula:

$$= IF(B2 >= 40, "Pass", "Fail")$$

Step 1:

$$50 >= 40$$

Step 2:
The condition is **TRUE**.
Step 3:
The TRUE result is:

Pass

COUNTIF vs IF

Function	Purpose
IF	Makes a decision for a cell
COUNTIF	Counts cells satisfying a condition

IF:

$$= IF(B2 >= 40, "Pass", "Fail")$$

Did this student pass?

COUNTIF:

$$= COUNTIF(C2 : C6, "Pass")$$

How many students passed?

Combined Spreadsheet Example

Student	Marks	Result
Arun	75	Pass
Anu	38	Fail
Rahul	65	Pass
Meera	82	Pass
Amal	45	Pass

We can use all the required functions:

- SUM $\rightarrow$ Total marks
- AVERAGE $\rightarrow$ Average marks
- COUNT $\rightarrow$ Number of marks
- COUNTIF $\rightarrow$ Number of passes
- IF $\rightarrow$ Pass or Fail

Combined Example – Formulas

Result in C2:

$$= IF(B2 >= 40, "Pass", "Fail")$$

Copy the formula down.

Total:

$$= SUM(B2 : B6)$$

$$75 + 38 + 65 + 82 + 45 = 305$$

Average:

$$= AVERAGE(B2 : B6) = 61$$

Combined Example – More Formulas

Number of students:

$$= COUNT(B2 : B6)$$

$$\boxed{5}$$

Number of students who passed:

$$= COUNTIF(C2 : C6, \text{"Pass"})$$

$$\boxed{4}$$

All five required functions

SUM AVERAGE COUNT COUNTIF IF

Cell referencing is used throughout the formulas.

Spreadsheet Classroom Activity

Enter the following data:

Student	Marks
A	35
B	55
C	72
D	40
E	28
F	90

Calculate:

1. Total marks using SUM
2. Average marks using AVERAGE
3. Number of students using COUNT
4. Result using IF
5. Number of students who passed using COUNTIF

Spreadsheet Quick Recap

Function	Use
SUM	Calculates total
AVERAGE	Calculates arithmetic mean
COUNT	Counts numeric cells
COUNTIF	Counts cells satisfying a condition
IF	Makes a decision based on a condition

Remember

$$\boxed{\text{Cell Reference} = \text{Column} + \text{Row}}$$

Presentation Design

What is a Presentation?

A **presentation** is a structured way of communicating information or ideas to an audience using slides.

Examples of presentation software:

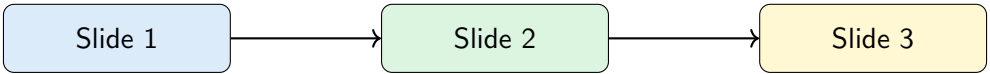
- Microsoft PowerPoint
- LibreOffice Impress

A presentation consists of multiple slides arranged in a logical order.

What is a Slide?

Definition

A **slide** is one page or screen of a digital presentation.



What is a Pitch Deck?

Simple Definition

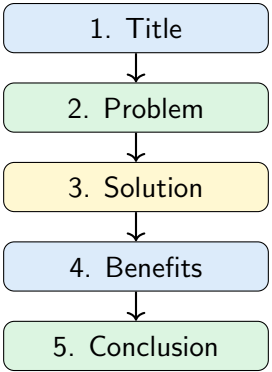
A **pitch deck** is a short presentation used to communicate an idea clearly and quickly to an audience.

Possible student topics:

- Smart Campus
- Digital Library
- Smart Parking
- Campus Recycling
- Student Study App

5-Slide Pitch Deck

The practical activity requires a **5-slide pitch deck**.



Slide 1 – Title

The first slide introduces the idea.
Include:

- Project / idea name
- Student name
- Roll number

Example

SMART CAMPUS

An Idea for a Better College

Name: Student Name
Roll No: 01

Slide 2 – Problem

Explain the problem that your idea addresses.

Example – Smart Parking

- Students spend time searching for parking.
- Available spaces are difficult to identify.
- This creates unnecessary waiting.

Tip:
Keep the information short and easy to read.

Slide 3 – Solution

Explain your proposed idea.

Example – Smart Parking

A smart parking system can help students identify available parking spaces more easily.

The audience should understand:

- What is the proposed idea?
- How does it address the problem?

Slide 4 – Benefits

Explain the main advantages of the idea.

Example

- Saves time
- Reduces unnecessary waiting
- Makes parking more organized
- Improves campus convenience

Tip:
Use short points instead of long paragraphs.

Slide 5 – Conclusion

The final slide summarizes the idea.

Example

Conclusion

Smart parking can make campus parking simpler, faster and more organized.

Thank You

Basic Presentation Design Principles

A good presentation should be:

Simple

Avoid unnecessary information.

Clear

The main idea should be easy to understand.

Readable

Use text that the audience can easily read.

Consistent

Maintain a consistent style.

Organized

Arrange information logically.

What is Animation?

Definition

An **animation** is a visual effect applied to an object on a slide.

Objects that can have animations include:

- Text
- Images
- Shapes
- Other slide objects

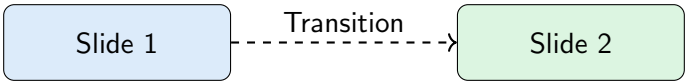
Example

Bullet points can be made to appear one after another during a presentation.

What is a Transition?

Definition

A **transition** is a visual effect that occurs when moving from one slide to another.



Animation vs Transition

Animation	**Transition**
Applied to objects on a slide	Applied when changing slides
Can affect text or images	Affects the change between slides
Example: text appears	Example: slide fades into the next slide

Easy Memory Trick

Animation = Object

Transition = Slide

Using Animations and Transitions Properly

Animations and transitions should:

- Support the presentation
- Keep the audience focused
- Make the presentation easier to follow

Avoid:

- Too many effects
- Distracting animations
- Unnecessary transitions

Remember

The purpose of a presentation is to communicate the idea clearly.

Classroom Activity – Create a 5-Slide Pitch Deck

Choose one simple idea:

- Smart Campus
- Digital Library
- Campus Recycling
- Smart Parking
- Student Study App

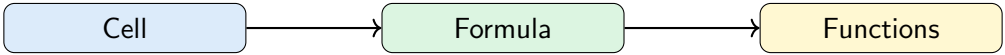
Create exactly **5 slides**:

1. Title
2. Problem
3. Solution
4. Benefits
5. Conclusion

Then:

- Apply basic formatting.
- Add a simple animation.

Final Recap – Spreadsheet



SUM	Total
AVERAGE	Arithmetic mean
COUNT	Number of numeric cells
COUNTIF	Count based on condition
IF	Decision based on condition

5-Slide Pitch Deck

1	Title
2	Problem
3	Solution
4	Benefits
5	Conclusion

Animation = Object

Transition = Slide

Quick Check

Spreadsheet

1. What is a cell address?
2. What does SUM do?
3. What is the purpose of AVERAGE?
4. What does COUNT count?
5. What is the difference between IF and COUNTIF?
6. What is cell referencing?

Presentation

1. What is a slide?
2. What is a pitch deck?
3. How many slides are required?
4. What is animation?
5. What is transition?

Thank You

Spreadsheet Basics and Presentation Design

Experiments 5 and 6